Java Programming: Software, JDK, JRE and IDEs

A beginner-friendly guide to the tools used to write, compile and run Java programs

1. Software That Helps to Run Java Programs

Java programs can be written and run using a text editor or an Integrated Development Environment (IDE), together with the Java Development Kit (JDK). Commonly used tools include:

- **JDK (Java Development Kit):** The essential software package for developing Java applications. It includes tools to compile and run Java code.
- **Eclipse IDE:** A popular, free IDE with a code editor, compiler integration, debugger and project-management tools.
- **Apache NetBeans:** A free IDE that supports Java development and provides project templates, code completion and debugging.
- **IntelliJ IDEA:** An IDE with a free Community Edition and a paid Ultimate Edition. It offers code assistance, debugging and project tools.
- **Visual Studio Code:** A lightweight code editor that can support Java through extensions. A JDK is still needed to compile and run Java programs.
- **Command Prompt / Terminal:** Java can also be compiled and run using JDK commands such as `javac` and `java`.

2. What Are JDK and JRE?

JDK (Java Development Kit): The JDK is a software development package used to create Java programs. It includes the Java compiler (`javac`), the Java launcher and other development tools, along with the runtime components needed to run Java applications.

JRE (Java Runtime Environment): The JRE traditionally refers to the environment required to run Java applications. It includes the Java Virtual Machine (JVM) and supporting libraries, but not the Java compiler. In modern Java releases, a separate public JRE download is not always provided; the JDK includes what is needed to run Java programs.

JDK	JRE
Used to develop, compile and run Java programs.	Traditionally used to run Java programs.
Includes development tools such as the compiler.	Does not include the Java compiler.
Required when writing and compiling Java code.	Provides the runtime environment for applications.

3. What Is Eclipse IDE?

Eclipse IDE (Integrated Development Environment) is a free, open-source software application mainly used for developing Java programs. It brings several programming tools together in one workspace, so developers do not need to use separate applications for every task.

- **Code editor:** Provides syntax highlighting, formatting and code suggestions.
- **Compilation support:** Detects many errors while writing code and builds Java projects.
- **Debugger:** Helps locate errors by running a program step by step.
- **Project management:** Organizes source files, libraries and project settings.

Official website: https://www.eclipse.org/ide/

4. How to Run a Java Program in Eclipse or NetBeans IDE

A. Running a Java Program in Eclipse

1 Download and install a suitable JDK. Then download and install Eclipse IDE for Java Developers.

2. Open Eclipse. Choose a workspace folder where your projects will be saved.
3. Select **File** → **New** → **Java Project**. Enter a project name and click **Finish**.
4. In the Package Explorer, right-click the **src** folder and select **New** → **Class**.
5. Enter a class name, for example **HelloWorld**. Select the option to create a `public static void main(String[] args)` method, then click **Finish**.
6. Type the Java code in the editor. Save the file using **Ctrl + S**.
7. Right-click the file and choose **Run As** → **Java Application**, or click the green Run button.
8. The program output appears in the Console panel at the bottom of the Eclipse window.

B. Running a Java Program in NetBeans

1. Download and install a suitable JDK and Apache NetBeans IDE.
2. Open NetBeans and select **File** → **New Project**.
3. Choose **Java** and a suitable Java application project template, then click **Next**.
4. Enter a project name, choose a project location and click **Finish**.
5. Open the generated main class, or right-click the source package and select **New** → **Java Class** to create one.
6. Write the Java code inside the main method and save the file.
7. Right-click the project and select **Run**, or press **F6** to run the main project.
8. View the program output in the Output window.

Example Java Program

```
public class HelloWorld {
public static void main(String[] args) {
System.out.println("Hello, World!");
}
}
```

Expected output: Hello, World!
Note: The class name and file name should match when the class is declared public. If the IDE asks for a JDK location, select the folder where the JDK is installed.

Conclusion: The JDK provides the tools needed to develop Java applications, while an IDE such as Eclipse or NetBeans makes writing, organizing, debugging and running programs easier.